

A Project Report on

**“A COMPARATIVE STUDY OF TRANSFER LEARNING MODELS
FOR WEED AND CROP CLASSIFICATION USING DEEP
LEARNING”**

Submitted in partial fulfilment for the award of the degree of

**BACHELOR OF TECHNOLOGY IN
COMPUTER SCIENCE AND ENGINEERING - GENERAL**

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ABSTRACT

Agriculture plays a vital role in global food production, and effective weed management is essential for improving crop productivity and ensuring sustainable farming practices. Weeds compete with crops for nutrients, water, sunlight, and space, resulting in reduced agricultural yield and increased economic loss. Traditional weed identification methods are manual, time-consuming, labor-intensive, and inefficient for large-scale farming environments. Therefore, there is a growing need for intelligent automated systems capable of accurately distinguishing weeds from crop plants in real time.

Recent advancements in Artificial Intelligence (AI), Deep Learning (DL), and Computer Vision have significantly improved agricultural image analysis and plant classification systems. Convolutional Neural Networks (CNNs) can automatically learn important visual features from images without requiring manual feature extraction. However, training deep neural networks from scratch requires large datasets, high computational resources, and longer training time. To overcome these challenges, transfer learning techniques are used, where pretrained models are fine-tuned on agricultural datasets to achieve high classification accuracy with reduced computational cost.

This project proposes a weed and crop classification system using transfer learning models for precision agriculture applications. The study evaluates three lightweight deep learning architectures namely MobileNetV1, MobileNetV2, and EfficientNetB0 using the V2 Plant Seedlings Dataset containing twelve different classes of crop and weed seedlings. The dataset is preprocessed using image resizing, normalization, and data augmentation techniques such as flipping, rotation, and zooming to improve model robustness and generalization capability. The dataset is divided into training, validation, and testing subsets using a 70:20:10 ratio.

A common classification head consisting of Global Average Pooling, Batch Normalization, Dropout, and Softmax layers is attached to each transfer learning model. The training process is carried out in two phases involving feature extraction and fine-tuning with reduced learning rates. Experimental results show that EfficientNetB0 achieved the highest classification accuracy of 98.05%, outperforming MobileNetV1 and MobileNetV2 in terms of accuracy, validation loss, and classification performance. The proposed system demonstrates the effectiveness of transfer learning in agricultural image classification and highlights its applications in automated weed detection, precision farming, selective herbicide spraying, and smart agricultural systems.

Chapter 1

Introduction

1.1 Background & Motivation

Agriculture is one of the most important sectors contributing to global food production and economic growth. With the increasing population, the demand for higher agricultural productivity has also increased significantly. One of the major challenges faced in agriculture is weed infestation, which negatively affects crop growth and reduces overall yield quality and quantity. Weeds compete with crops for essential resources such as nutrients, sunlight, water, and root space, leading to considerable agricultural losses every year. Traditional weed management methods mainly involve manual inspection and herbicide spraying, which are time-consuming, labor-intensive, expensive, and unsuitable for large-scale farming.

Excessive use of herbicides also causes environmental pollution, soil degradation, contamination of water resources, and the development of herbicide-resistant weed species. Therefore, there is a growing need for intelligent automated systems capable of accurately distinguishing weeds from crop plants in real time. Recent advancements in Artificial Intelligence (AI), Deep Learning (DL), and Computer Vision technologies have transformed agricultural applications by enabling automated plant recognition and classification systems. Deep learning models, especially Convolutional Neural Networks (CNNs), can automatically learn important image features such as shape, texture, color, and patterns without requiring manual feature extraction.

However, training deep learning models from scratch requires large datasets, high computational resources, and longer training times. Agricultural datasets are often limited in size, making model training difficult. Transfer learning addresses this problem by using pretrained models that have already learned feature representations from large datasets such as ImageNet. These pretrained models can then be fine-tuned on agricultural datasets with reduced computational cost and improved performance. This project focuses on developing an automated weed and crop classification system using transfer learning techniques.

Three lightweight deep learning architectures namely MobileNetV1, MobileNetV2, and EfficientNetB0 are evaluated using the V2 Plant Seedlings Dataset containing twelve classes of weed and crop seedlings. The proposed system applies preprocessing and data augmentation techniques such as image resizing, normalization, flipping, rotation, and zooming to improve robustness and reduce overfitting. The transfer learning models are trained using feature extraction and fine-tuning processes. Experimental results demonstrate that EfficientNetB0 achieves superior classification performance compared to MobileNetV1 and MobileNetV2, highlighting the effectiveness of transfer learning for precision agriculture applications such as automated weed detection and smart farming systems.

1.2 Objectives

- The primary objectives of the proposed weed and crop classification system are as follows:
- To develop an intelligent weed and crop classification system using transfer learning and deep learning techniques.
- To evaluate the performance of MobileNetV1, MobileNetV2, and EfficientNetB0 architectures for agricultural image classification.
- To apply image preprocessing and data augmentation techniques for improving model robustness and generalization capability.
- To reduce computational complexity and training time using pretrained deep learning models.
- To classify twelve different weed and crop seedling classes using the V2 Plant Seedlings Dataset.

1.3 Benefits of Research

The proposed weed and crop classification system provides several important benefits for agriculture and precision farming applications.

- Farmers can reduce manual labor involved in weed identification and monitoring processes.
- Accurate weed detection helps in reducing unnecessary herbicide usage, thereby minimizing environmental pollution and soil damage.
- Automated classification systems improve crop productivity by enabling early weed management.
- Transfer learning reduces computational cost and training time while achieving high classification accuracy.
- The lightweight deep learning models used in this project are suitable for deployment on mobile devices, drones, and agricultural robots.
- Precision agriculture systems based on deep learning can improve farming efficiency and decision-making processes.
- The proposed system supports sustainable agricultural practices by reducing chemical overuse and improving resource management.
- The research contributes to the advancement of AI-based agricultural technologies and smart farming applications

Chapter 2

Literature Survey

2.1 Literature Review

A comprehensive review of existing literature was conducted to understand the current state of deep learning and transfer learning techniques used for weed and crop classification systems. The review helped identify existing challenges, research gaps, and suitable methodologies for developing an intelligent agricultural image classification system. The following table summarizes some important research works related to the proposed study.

Author(s) & Year	Title	Methodology	Findings	Limitations
Jose et al. (2022)	Classification of Weeds and Crops using Transfer Learning	Transfer Learning with CNNs	Improved weed classification accuracy using pretrained models	Limited dataset size
Vypirailenko et al. (2021)	Deep Learning Techniques for Weed Growth Classification	Deep Learning Models	Accurate weed growth-stage classification	High computational complexity
Espejo-Garcia et al. (2020)	Transfer Learning for Weed Identification Assistance	MobileNet, DenseNet, Xception	Better classification compared to traditional methods	Requires high-quality datasets
Howard et al. (2017)	MobileNetV1 Architecture	Depthwise Separable Convolutions	Reduced parameters and computational cost	Lower accuracy than deeper networks
Sandler et al. (2018)	MobileNetV2 Architecture	Inverted Residual Blocks	Improved feature extraction efficiency	Moderate classification performance
Tan and Le (2019)	EfficientNet Architecture	Compound Scaling Method	High accuracy with fewer parameters	Slightly complex architecture

2.1.1 Analysis of Literature

The literature review reveals that deep learning and transfer learning techniques have significantly improved agricultural image classification systems compared to traditional machine learning approaches. Transfer learning models pretrained on large datasets such as ImageNet are capable of learning rich feature representations that can be effectively applied to agricultural datasets with limited training samples.

Several researchers have demonstrated that Convolutional Neural Networks (CNNs) can accurately classify weeds and crops based on image features such as texture, color, shape, and plant morphology. Lightweight architectures such as MobileNetV1 and MobileNetV2 are particularly suitable for mobile and embedded agricultural devices because of their low computational requirements and reduced model size.

EfficientNet models introduced improved scaling strategies that balance network depth, width, and input resolution to achieve better classification accuracy with fewer parameters. Among the reviewed architectures, EfficientNet demonstrated superior performance and convergence stability in image classification tasks.

The literature also highlights the importance of preprocessing and data augmentation techniques such as image normalization, flipping, rotation, and zooming. These techniques improve model robustness, reduce overfitting, and increase classification accuracy under real-world agricultural conditions.

Although several studies have focused on weed classification using deep learning, only limited research has directly compared MobileNetV1, MobileNetV2, and EfficientNetB0 using the same dataset, preprocessing pipeline, augmentation strategy, and evaluation metrics. Therefore, this project aims to fill this research gap by performing a comparative analysis of these lightweight transfer learning architectures for weed and crop classification using the V2 Plant Seedlings Dataset.

2.2 Problem Formulation and Proposed Work

2.2.1 Introduction

Weed and crop classification is an important task in precision agriculture because weeds reduce crop productivity by competing for nutrients, water, and sunlight. Traditional weed detection methods are manual, time-consuming, and inefficient for large-scale farming. Recent advancements in Artificial Intelligence (AI) and Deep Learning (DL) have enabled automated plant classification systems using image analysis. Transfer learning models provide better accuracy with lower computational cost and training time. Therefore, this project focuses on developing an automated weed and crop classification system using lightweight transfer learning architectures.

2.2.2 Problem Statement

Traditional weed management techniques require manual monitoring and excessive herbicide usage, which increases labor cost and environmental pollution. Existing image classification systems often suffer from limited accuracy, high computational complexity, and poor real-world performance. Many models are difficult to deploy on mobile and edge devices due to large model size. Hence, there is a need for an efficient and lightweight weed classification system capable of accurately identifying weeds and crops. The proposed system aims to solve this problem using transfer learning and deep learning techniques.

2.2.3 System Architecture / Model

The proposed system uses a transfer learning-based deep learning architecture for weed and crop classification. Input images from the V2 Plant Seedlings Dataset are preprocessed using resizing and normalization techniques. Data augmentation methods such as flipping, rotation, and zooming are applied to improve model robustness. MobileNetV1, MobileNetV2, and EfficientNetB0 are used as pretrained feature extraction models. A classification head consisting of pooling, normalization, dropout, and softmax layers is added for final classification.

2.2.4 Proposed Algorithms

The proposed system uses three lightweight transfer learning architectures namely MobileNetV1, MobileNetV2, and EfficientNetB0. MobileNetV1 uses depthwise separable convolutions to reduce computational complexity. MobileNetV2 improves feature extraction using inverted residual blocks and linear bottlenecks. EfficientNetB0 uses compound scaling techniques to achieve better accuracy and efficiency. Data augmentation algorithms are also applied to improve generalization and reduce overfitting during training.

2.2.5 Proposed Work

The proposed work involves developing an automated weed and crop classification system using transfer learning techniques. The V2 Plant Seedlings Dataset containing twelve plant classes is used for experimentation. Preprocessing and augmentation techniques are applied before training the models. The performance of MobileNetV1, MobileNetV2, and EfficientNetB0 is compared using evaluation metrics such as accuracy, precision, recall, and F1-score. The best-performing model is identified for precision agriculture applications such as smart farming and automated weed detection systems.

Chapter 3

System Analysis and Proposed Methodology

3.1 Introduction

This chapter explains the system analysis and methodology used for the proposed weed and crop classification system. The system uses deep learning and transfer learning techniques to automatically classify weeds and crop seedlings from images. The methodology includes dataset preprocessing, image augmentation, model training, fine-tuning, and performance evaluation. Lightweight pretrained CNN architectures are used to improve classification accuracy with lower computational cost. The proposed system is designed to support precision agriculture and smart farming applications.

3.2 Problem Statement

Weed infestation is one of the major problems affecting agricultural productivity and crop quality. Traditional weed identification methods require manual monitoring and herbicide spraying, which are time-consuming, labor-intensive, and environmentally harmful. Existing automated systems often suffer from limited accuracy and high computational complexity. Many models are difficult to deploy on mobile and embedded agricultural devices because of large model size and resource requirements. Therefore, an efficient and lightweight weed classification system is needed for real-time agricultural applications.

3.3 System Architecture / Model

The proposed system follows a transfer learning-based deep learning architecture for weed and crop classification. Images from the V2 Plant Seedlings Dataset are first preprocessed using resizing and normalization techniques. Data augmentation methods such as horizontal flipping, vertical flipping, rotation, and zooming are applied to improve robustness and reduce overfitting. MobileNetV1, MobileNetV2, and EfficientNetB0 are used as pretrained feature extraction models. A classification head containing pooling, normalization, dropout, and softmax layers is added for final plant classification.

3.4 Proposed Algorithms

The proposed system uses transfer learning algorithms based on MobileNetV1, MobileNetV2, and EfficientNetB0 architectures. MobileNetV1 reduces computational complexity using depthwise separable convolutions. MobileNetV2 improves feature extraction using inverted residual blocks and linear bottleneck layers. EfficientNetB0 uses compound scaling techniques to balance network depth, width, and image resolution for better accuracy. Data augmentation algorithms are also applied to improve generalization and reduce overfitting during training.

3.5 Proposed Work

The proposed work involves developing an automated weed and crop classification system using transfer learning models. The V2 Plant Seedlings Dataset containing twelve plant classes is used for experimentation. The dataset is preprocessed and augmented before model training. MobileNetV1, MobileNetV2, and EfficientNetB0 are trained and evaluated using the same preprocessing and training conditions. The performance of the models is analyzed using evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrix analysis. The best-performing model is selected for precision agriculture applications such as smart farming and automated weed detection systems.

Chapter 4

System Design and Implementation

4.1 Introduction

This chapter explains the design and implementation of the proposed weed and crop classification system. The system is developed using deep learning and transfer learning techniques for automated plant classification. The implementation includes dataset preprocessing, data augmentation, transfer learning model integration, training, and performance evaluation. Lightweight CNN architectures are used to achieve high classification accuracy with reduced computational cost. The system is designed for precision agriculture applications such as smart farming and automated weed detection.

4.2 System Design Principles

The proposed system is designed based on important deep learning and image classification principles. The system uses transfer learning to reduce training time and improve classification accuracy. Data preprocessing and augmentation techniques are applied to improve robustness and reduce overfitting. Lightweight architectures such as MobileNetV1 and MobileNetV2 are selected for low computational complexity, while EfficientNetB0 is used for improved feature extraction and accuracy. The system also follows a modular architecture for easier implementation and performance comparison.

4.3 System Architecture

The proposed architecture consists of image preprocessing, augmentation, feature extraction, classification, and evaluation stages. Input images from the V2 Plant Seedlings Dataset are resized and normalized before training. Augmented images are then passed to pretrained transfer learning models namely MobileNetV1, MobileNetV2, and EfficientNetB0. A common classification head containing Global Average Pooling, Batch Normalization, Dropout, and Softmax layers is attached to each model. Finally, the system evaluates classification performance using multiple evaluation metrics.

4.4 Dataset Design & Preprocessing

The system uses the V2 Plant Seedlings Dataset containing twelve classes of weeds and crop seedlings. The dataset is divided into training, validation, and testing subsets in the ratio of 70:20:10. Image preprocessing techniques such as resizing and normalization are applied before model training. Data augmentation methods including horizontal flipping, vertical flipping, rotation, and zooming are used to improve model generalization and reduce overfitting. These preprocessing steps improve the robustness and efficiency of the classification system.

4.5 Model Architecture & Implementation

The proposed system evaluates three transfer learning architectures namely MobileNetV1, MobileNetV2, and EfficientNetB0. MobileNetV1 uses depthwise separable convolutions to reduce computational cost. MobileNetV2 improves feature extraction using inverted residual blocks and linear bottleneck layers. EfficientNetB0 uses compound scaling techniques to balance network depth, width, and resolution for better accuracy. The models are initialized using pretrained ImageNet weights and trained using a two-stage transfer learning process involving feature extraction and fine-tuning.

4.6 Training and Evaluation

The training process is performed in two phases. In the first phase, the pretrained backbone layers are frozen and only the classification layers are trained. In the second phase, the complete model is fine-tuned using a lower learning rate to improve feature adaptation. The models are evaluated using performance metrics such as accuracy, precision, recall, F1-score, validation loss, and confusion matrix analysis. Experimental results are compared to identify the best-performing architecture for weed and crop classification.

4.7 Implementation Workflow

The implementation workflow begins with dataset collection and preprocessing. The images are resized, normalized, and augmented before training. Pretrained transfer learning models are then initialized and connected with a classification head. The models are trained and fine-tuned using the prepared dataset. Finally, the system evaluates the classification performance and compares the results of MobileNetV1, MobileNetV2, and EfficientNetB0. The best-performing model is selected for precision agriculture applications such as smart farming and automated weed detection system

Chapter 5

Tools and Technologies Required

5.1 Development Tools

The proposed weed and crop classification system is developed using various software tools and deep learning frameworks. Python is used as the primary programming language because of its simplicity and extensive support for machine learning libraries. Google Colab is used as the development and training environment since it provides free GPU support for deep learning model training. Jupyter Notebook is also used for experimentation, preprocessing, and visualization tasks. Visual Studio Code is used as the code editor for implementation and debugging purposes.

5.2 Frontend Technologies

The proposed system mainly focuses on deep learning-based image classification and visualization. Matplotlib and Seaborn libraries are used for plotting graphs such as accuracy curves, loss curves, and confusion matrices. OpenCV is used for image processing and augmentation operations including resizing, rotation, flipping, and normalization. These technologies help in visualizing model performance and improving the quality of input images used for training.

Technology	Category	Purpose
Matplotlib	Visualization Tool	Plotting accuracy and loss graphs
Seaborn	Visualization Library	Confusion matrix visualization
OpenCV	Image Processing	Image preprocessing and augmentation

5.3 Backend Technologies

TensorFlow and Keras are used as the primary deep learning frameworks for implementing transfer learning models. NumPy is used for numerical computations and handling multidimensional arrays. Pandas is used for dataset handling and data analysis tasks. The pretrained architectures namely MobileNetV1, MobileNetV2, and EfficientNetB0 are implemented using TensorFlow-Keras libraries. These frameworks provide efficient model training, evaluation, and fine-tuning capabilities for deep learning applications.

Technology	Category	Purpose
Python	Programming Language	Model implementation
TensorFlow	Deep Learning Framework	Model training and evaluation
Keras	Deep Learning API	Transfer learning implementation
NumPy	Numerical Library	Array and matrix operations
Pandas	Data Analysis	Dataset management

5.4 Deployment & Hosting Tools

Google Colab is used for training and testing the deep learning models because it provides cloud-based GPU support and faster computation. The trained models can be deployed on edge devices, mobile systems, agricultural robots, and smart farming applications. The lightweight architectures such as MobileNetV1 and

MobileNetV2 are suitable for deployment on resource-constrained devices due to their lower computational requirements. EfficientNetB0 can also be integrated into real-time precision agriculture systems for automated weed detection and classification

Chapter 6

Software Requirement Specification (SRS)

6.1 Specific Requirements

The proposed weed and crop classification system is designed to automatically identify and classify weeds and crop seedlings using deep learning and transfer learning techniques. The system requires image preprocessing, data augmentation, feature extraction, model training, and evaluation functionalities. It should support lightweight deep learning architectures capable of providing high classification accuracy with reduced computational complexity. The system must also be capable of handling agricultural image datasets efficiently for precision farming applications.

6.2 Functional Requirements

The functional requirements define the major operations performed by the proposed system. The system should accept plant seedling images as input and preprocess them using resizing and normalization techniques. Data augmentation methods such as flipping, rotation, and zooming must be applied to improve model robustness. The system should train and evaluate MobileNetV1, MobileNetV2, and EfficientNetB0 models using transfer learning techniques. It should also generate evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrix analysis for performance comparison.

Req. ID	Requirement	Priority	Description
FR-01	Image Preprocessing	High	Resize and normalize input images
FR-02	Data Augmentation	High	Apply flipping, rotation, and zooming
FR-03	Transfer Learning	High	Train pretrained CNN architectures
FR-04	Model Evaluation	High	Generate accuracy and loss metrics
FR-05	Performance Comparison	Medium	Compare MobileNet and EfficientNet models
FR-06	Confusion Matrix Analysis	Medium	Analyze classification results

6.3 Non-Functional Requirements

The system should provide high classification accuracy with minimal validation loss. It should support efficient training and testing with low computational requirements. The lightweight architectures should enable deployment on mobile and edge-based agricultural devices. The system should be scalable for larger agricultural datasets and capable of handling real-world farming conditions. It should also provide reliable and robust performance under different image variations and environmental conditions.

6.4 System Interfaces

The system uses image datasets as input and produces classification results as output. TensorFlow and Keras frameworks are used for implementing the deep learning models. OpenCV is used for image preprocessing and augmentation operations. Matplotlib and Seaborn are used for visualizing performance graphs and confusion matrices. The system can be integrated with agricultural robots, drones, and smart farming systems for automated weed detection and precision agriculture applications.

Chapter 7

Results and Discussion

7.1 Introduction

This chapter presents the results obtained from the implementation and evaluation of the proposed weed and crop classification system. The performance of three transfer learning architectures namely MobileNetV1, MobileNetV2, and EfficientNetB0 is analyzed using the V2 Plant Seedlings Dataset. The models are evaluated using performance metrics such as accuracy, precision, recall, F1-score, validation loss, and confusion matrix analysis. The objective of this chapter is to compare the classification performance of the models and identify the most effective architecture for precision agriculture applications.

7.2 Model-wise Results

The experimental results show significant differences in the performance of the evaluated transfer learning models. MobileNetV1 achieved a test accuracy of 87.79% with satisfactory classification performance and lower computational complexity. MobileNetV2 achieved an accuracy of 81.42%, but its validation accuracy fluctuated during training, resulting in comparatively lower performance. EfficientNetB0 achieved the highest test accuracy of 98.05% with minimal validation loss and better convergence stability.

The results indicate that EfficientNetB0 provides superior feature extraction and classification capability compared to MobileNetV1 and MobileNetV2. The model successfully classified most weed and crop classes with very few misclassification errors. The lightweight nature of MobileNet architectures still makes them suitable for mobile and edge-based agricultural devices where computational resources are limited.

7.3 Performance Metrics

The performance of the transfer learning models is evaluated using various classification metrics. Accuracy measures the overall classification performance of the model, while precision, recall, and F1-score evaluate the quality of predictions for individual classes. Validation loss is used to analyze model convergence and overfitting behavior during training.

Model	Test Accuracy (%)	Test Loss
MobileNetV1	87.79	0.3422
MobileNetV2	81.42	0.5843
EfficientNetB0	98.05	0.0868

The results clearly show that EfficientNetB0 outperformed the other models in terms of classification accuracy and loss reduction. The lower validation loss indicates better generalization and feature extraction capability.

7.4 Discussion of Findings

The experimental findings demonstrate that transfer learning is highly effective for agricultural image classification tasks. EfficientNetB0 achieved the best performance because of its compound scaling strategy, which balances network depth, width, and image resolution efficiently. MobileNetV1 also achieved good performance with lower computational complexity, making it suitable for deployment on embedded agricultural systems.

The study also highlights the importance of preprocessing and data augmentation techniques in improving model robustness and reducing overfitting. Techniques such as flipping, rotation, zooming, and normalization helped the models generalize better under varying image conditions. Overall, the proposed system demonstrates the potential of deep learning and transfer learning techniques for precision agriculture applications such as automated weed detection, smart herbicide spraying, and agricultural robotics.

Chapter 8

Conclusions and Future Scope

8.1 Conclusion

The proposed weed and crop classification system successfully demonstrates the effectiveness of transfer learning and deep learning techniques in precision agriculture applications. The project evaluated three lightweight transfer learning architectures namely MobileNetV1, MobileNetV2, and EfficientNetB0 using the V2 Plant Seedlings Dataset containing twelve plant classes. The models were trained and tested under identical preprocessing, augmentation, and training conditions to ensure fair performance comparison.

Experimental results showed that EfficientNetB0 achieved the highest classification accuracy of 98.05% with minimal validation loss, outperforming MobileNetV1 and MobileNetV2. The study proved that transfer learning can significantly improve agricultural image classification performance while reducing computational complexity and training time. The use of preprocessing and data augmentation techniques further improved model robustness and reduced overfitting.

The proposed system can support precision agriculture applications such as automated weed detection, selective herbicide spraying, smart farming systems, and agricultural robots. Overall, the project demonstrates that deep learning-based weed classification systems can improve farming efficiency, reduce manual effort, minimize herbicide misuse, and support sustainable agricultural practices.

8.2 Future Scope

Several improvements and future enhancements can be implemented to further improve the proposed system. More advanced deep learning architectures such as EfficientNetB4, ResNet50, Vision Transformers (ViT), and DenseNet can be evaluated using larger agricultural datasets for improved performance. Real-time weed detection systems can also be developed by integrating the trained models with drones, cameras, and agricultural robots.

The system can be extended for real-time field monitoring and automated herbicide spraying applications. Integration with IoT devices and smart farming systems can further improve agricultural automation and decision-making processes. Mobile and edge-device deployment can also be explored for low-cost and portable weed detection systems suitable for rural farming environments.

8.3 Summary

This project focused on the development of an automated weed and crop classification system using transfer learning techniques. Three lightweight deep learning architectures namely MobileNetV1, MobileNetV2, and EfficientNetB0 were evaluated using the V2 Plant Seedlings Dataset. The system applied preprocessing and data augmentation techniques to improve classification accuracy and model robustness.

Among the evaluated models, EfficientNetB0 achieved the best classification performance with 98.05% accuracy. The results confirmed that transfer learning is highly effective for agricultural image classification tasks. The proposed system demonstrates strong potential for precision agriculture applications including smart farming, automated weed detection, selective herbicide spraying, and agricultural robotics.

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